



Speaker Notes (Slide-by-Slide)

Slide 1: Introduction — Why This Topic Matters

"Today's AI systems aren't just made of data and algorithms—they're shaped by the minds of the people who build them. Developers use their own mental models—cognitive frameworks—to design how AI thinks. So, understanding developer thinking is crucial if we want to understand, improve, and even trust AI."

Slide 2: What Are Mental Models?

"In psychology, mental models are internal representations that help us understand, predict, and navigate the world. They let us simulate how things work in our minds before taking real actions. Developers also use them unconsciously—especially when building AI systems."

Slide 3: Mental Models in the Mind of a Developer

"A developer rarely starts by typing code. First, they imagine the logic: what inputs go in, how the system behaves, and what output comes out. This process is cognitive simulation—it's a mental model in action."

Slide 4: Developers Don't Code—They Think in Abstractions

"Instead of concrete actions, developers think in layers and patterns. Abstractions like 'If X then Y' or 'This class talks to that one' become the mental scaffolding behind the actual AI. So, the AI doesn't just reflect reality—it reflects how the developer *sees* reality."

Slide 5: Flawed Mental Models → Flawed AI

"If a developer misunderstands human behavior or has a biased view of intelligence, the AI inherits that. That's why facial recognition failed for

darker skin tones—it reflected biased data *and* biased thinking during design."

Slide 6: Feedback Loops in AI Design

"Once an AI system is live, developers get feedback. They tweak and improve it. Over time, their mental models evolve with the AI itself. But if their assumptions remain unchallenged, they can just reinforce bad habits."

Slide 7: Cognitive Biases in Developer Thinking

"Developers are human too—they fall prey to cognitive biases like confirmation bias (favoring data that proves their idea), or overconfidence. These biases sneak into AI systems and affect fairness, safety, and accuracy."

Slide 8: Misalignment Between Developer & User Mental Models

"Imagine using an app and it behaves in a way you don't expect. That disconnect? It's a mismatch between your mental model and the developer's. These gaps lead to frustration, misuse, or even dangerous errors."

Slide 9: Explainable AI (XAI) – A Bridge Between Minds

"Explainable AI tries to make AI's reasoning more understandable. If users can grasp why a model made a decision, they can align their mental model with the AI's logic. This boosts transparency and trust."

Slide 10: Tools to Build Better Developer Mental Models

"Tools like TensorBoard help visualize how neural networks learn. Collaborative tools like code reviews help expose flawed mental models. So, it's not just about coding—it's about seeing and questioning *how* we think."

Slide 11: Shared Mental Models in Team Environments

"In team settings, everyone brings a different model. If these aren't aligned, the final system becomes chaotic or inconsistent. Shared understanding is a psychological factor crucial for successful AI development."

Slide 12: Case Study — Self-Driving Cars

"Self-driving car developers model human perception—how we notice signs, avoid obstacles. But several accidents happened because developers *underestimated* edge cases or over-simplified real-world behavior."

Slide 13: The Future — Can AI Understand Our Mental Models?

"Imagine AI that adapts to *your* thinking. AI tutors adjusting to your learning style. Mental health apps detecting your emotions and suggesting support. The next big leap in AI might not be stronger intelligence—it might be *empathy*."

Slide 14: Thought-Provoking Questions

- Are we building machines—or copies of our cognitive biases?
- What would happen if an AI could *disagree* with a developer's flawed model?
- Could psychology help us create more ethical AI?

"Let's not just build smarter machines. Let's build *wiser* ones."

Slide 15: Conclusion

"Developers' mental models shape the way AI thinks, behaves, and interacts with us. Psychology gives us a lens to study and improve those models. The future of AI isn't just about technology—it's about understanding the minds behind it."

Engaging Class Activity: "AI as Mirror of the Mind"

 **Goal:** Show how our mental models shape intelligent systems—even in simple tasks.

Materials Needed:

- Whiteboard or chart paper
- Markers
- (Optional) Slips of paper or index cards